Grade 6 Accelerated — Name_Answer Key_________
Unit 15 — Date____________________
Lesson 4 Practice — Period___________________

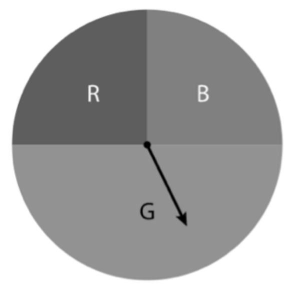
1. The probabilities of four events are shown. List them in order from least to greatest.

$\frac{6}{13}$ — 3 out of 9 — 41% — 0.2

0.2, 3 out of 9, 41%, $\frac{6}{13}$

Use the spinner to determine missing events, probabilities, and descriptions of likelihood. Express all probabilities as percentages. Use the vocabulary to complete each sentence.

impossible — unlikely — equally likely as unlikely — likely — certain



2. The probability of landing on green, $P(G)$, is __50%__. This event is __equally likely as unlikely__.

3. The probability of landing on red, $P(R)$, is __25%__. This event is __unlikely__.

4. The probability of landing on yellow, $P(Y)$, is __0%__. This event is __impossible__.

5. The probability of landing on red or blue, $P(\text{R or B})$, is __50%__. This event is __equally likely as unlikely__.

6. The probability of landing on a color, $P(\text{color})$ is __100%__. This event is __certain__.

Here are some scenarios for questions 7-8.

- According to market research, a business has a 75% chance of making money in the first 3 years. 75%
- According to lab testing, $\frac{5}{6}$ of a certain kind of experimental light bulb will work after 3 years. $83\frac{1}{3}\%$
- According to experts, the likelihood of a car needing major repairs in the first 3 years is 0.7. 70%

7. Write the scenarios in order of likelihood from least to greatest after three years: the business makes money, the light bulb still works, and the car needs major repairs.

   Car needs major repairs, business makes money, light bulb still works.

8. Name another chance experiment that has the same likelihood as one of the scenarios.

   Answers will vary.
   Ex: Using the spinner on the previous page, the likelihood of spinning and landing on green or red is 75%.

9. Which event is more likely: rolling a standard number cube and getting an even number, or flipping a coin and having it land heads up?

   Since the probability of either event happening is 50%, both are equally likely.

10. Miami International airport reported that on a given day, the probability that an airline arrived on time was 0.32. The likelihood that the next airplane arrived earlier than scheduled was $\frac{3}{10}$ and the later than scheduled was 37.8%. What is most likely to occur when the next airplane arrives?
    On time: 0.320
    Early: 0.300
    Late: 0.378

    Because its probability is largest, it is most likely that the plane will arrive later than scheduled.